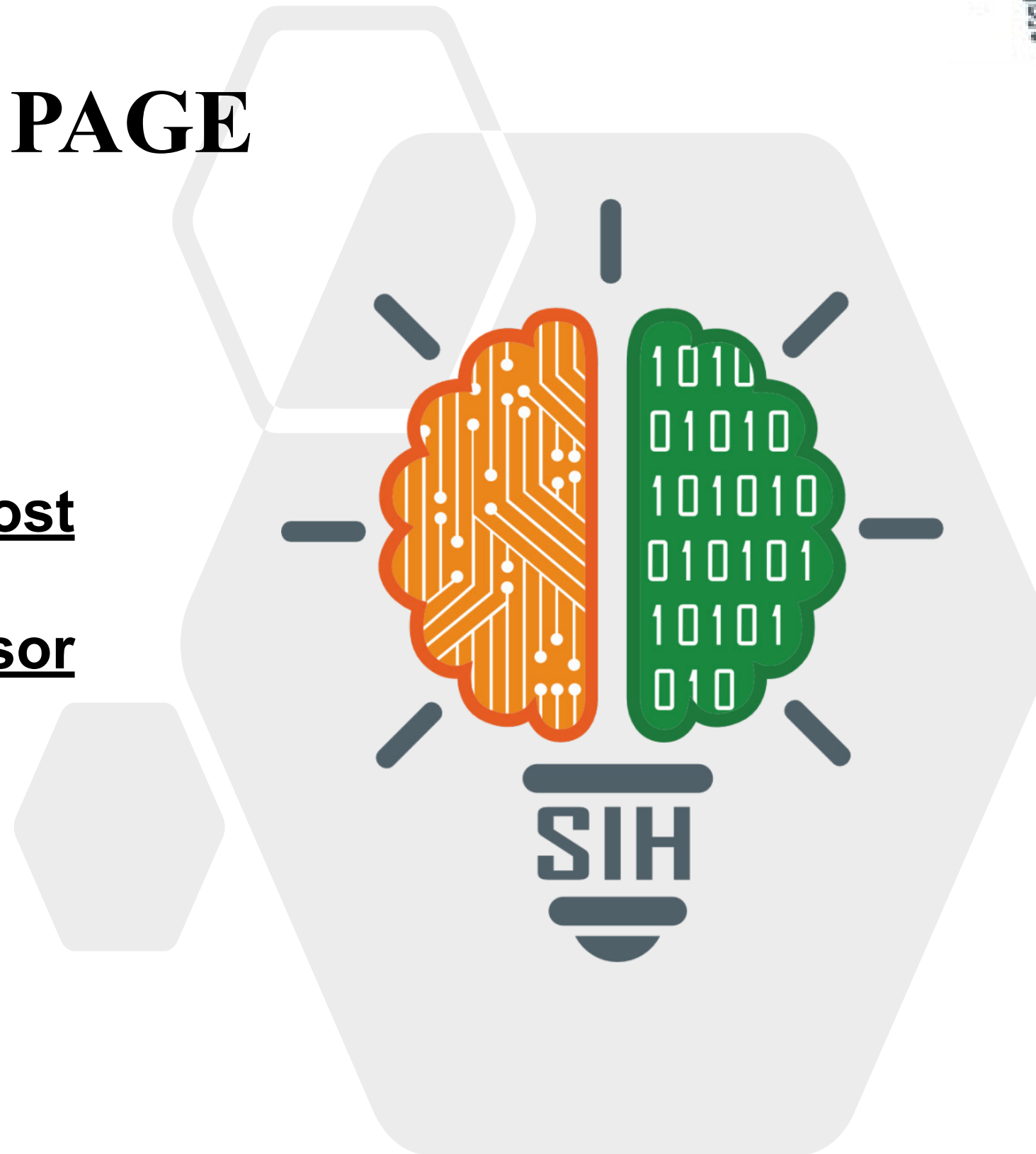
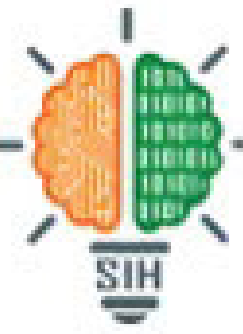


TITLE PAGE

- Problem Statement ID – SIH26064
- Problem Statement Title - Low-Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration
- Theme- Robotics and Drones
- PS Category- Hardware
- Team Name (Registered on portal) - EmbedX

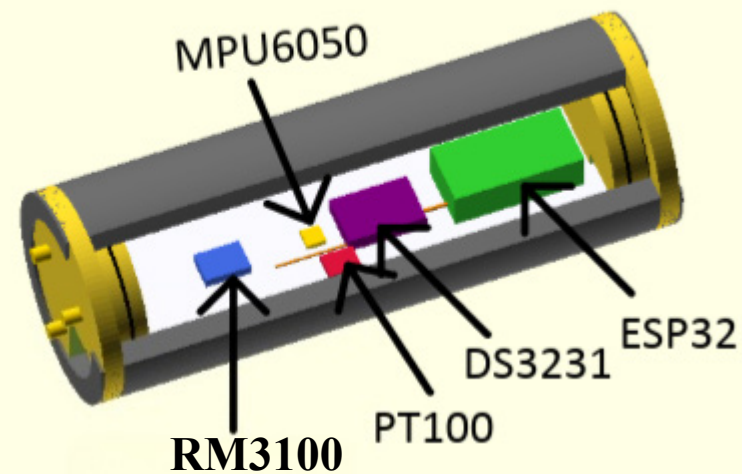


Low-Cost Deployable Seafloor Metal Detection & Localization



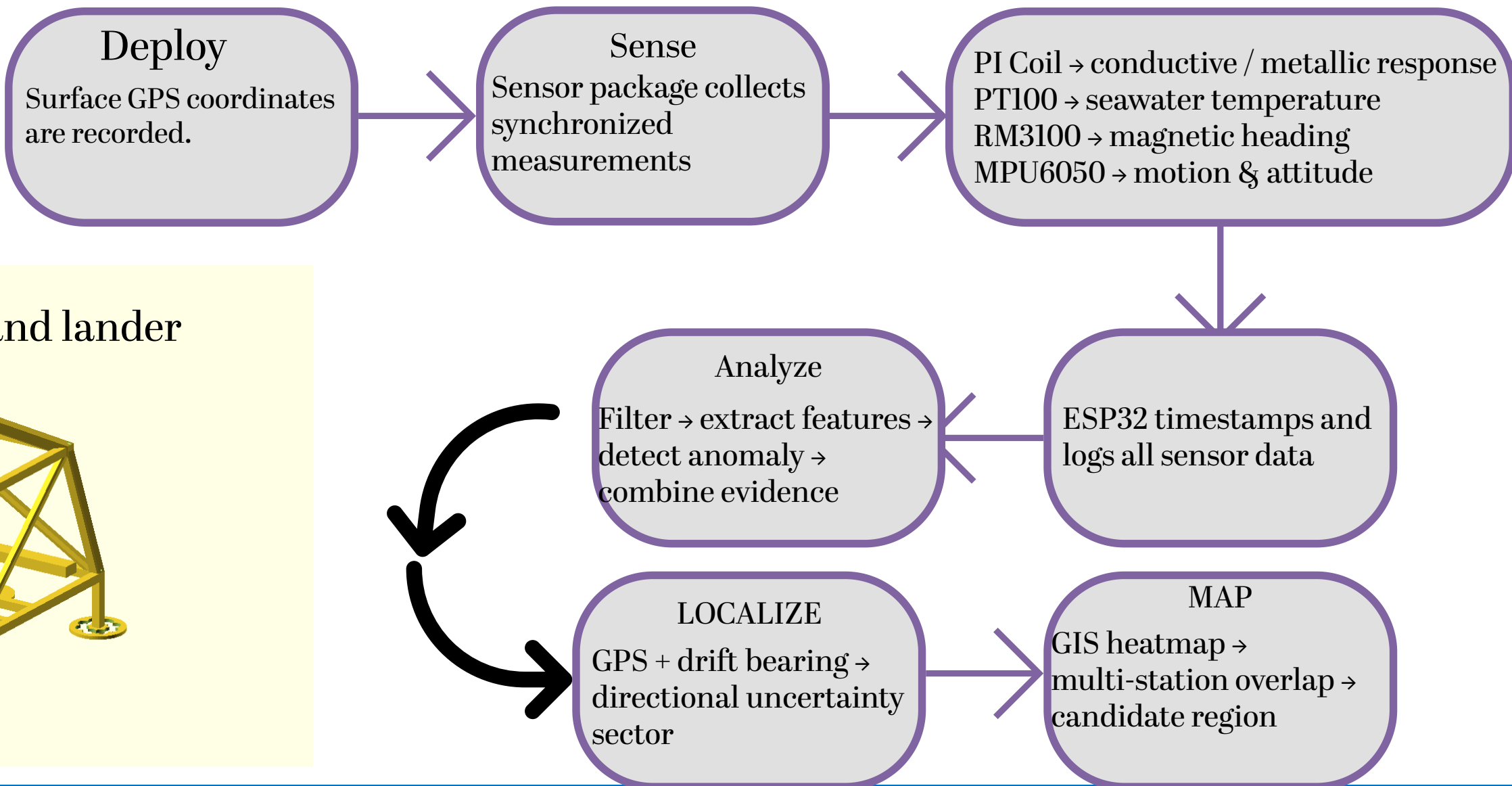
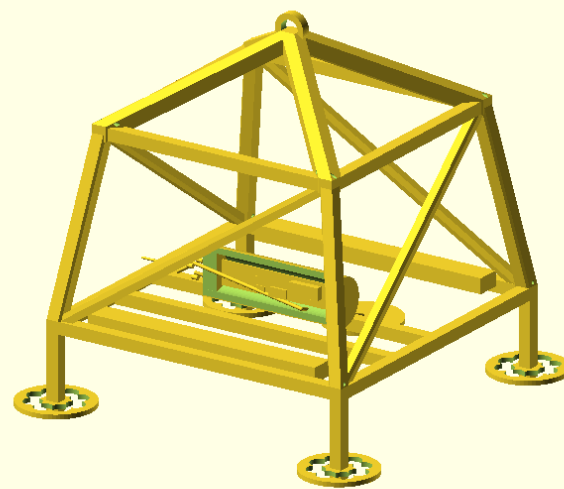
Proposed Solution:

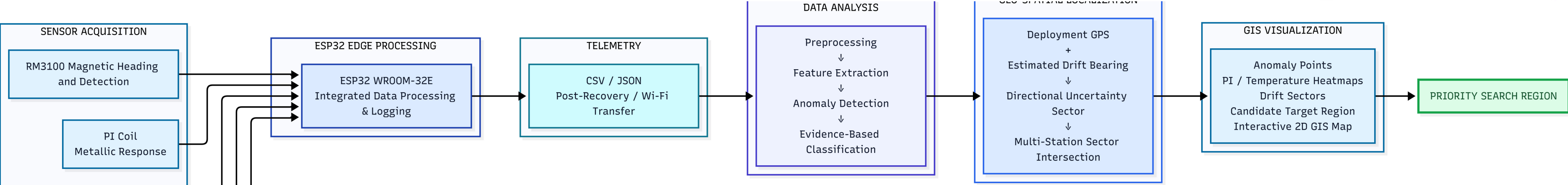
A compact deployable package combining **PI metal sensing**, **seawater temperature**, **magnetic heading** and **motion/attitude** to detect and prioritize seabed anomalies.



ESP32 synchronizes and logs sensor data for post-recovery analysis and GIS mapping.

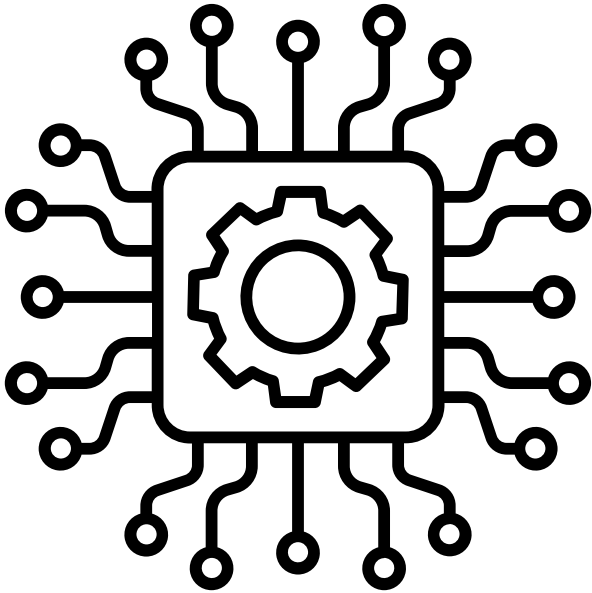
Payload carrier and lander





HOW THE SYSTEM WORKS

- Sensor Acquisition: Multi-sensor suite captures magnetic, metallic, temperature, motion and timestamp data.
- ESP32 Edge Processing: ESP32 collects, synchronizes and locally logs the raw sensor readings during deployment.
- Telemetry: After recovery, logged data is transferred via Wi-Fi in CSV/JSON format for processing.
- Data Analysis: Raw signals undergo preprocessing → feature extraction → anomaly detection → evidence-based classification.
- Geo-Spatial Localization: Magnetic heading, IMU motion data and current-aligned fins are fused to estimate the probable drift direction and uncertainty.
- GIS Visualization: Detected anomalies are mapped with heatmaps and directional uncertainty sectors and overlaid with available seafloor topography.
- Priority Search Region: Intersecting information from multiple deployments narrows the probable target area, guiding efficient ROV/field investigation.



TECHNICALLY FEASIBLE

- Low-cost COTS sensors: ESP32, MPU6050, RM3100, PT100 and RTC.
- Local data logging: Sensor operates underwater without requiring continuous communication.
- Post-recovery Wi-Fi transfer: Data can be uploaded after retrieval.
- Pressure-resistant torpedo-style housing enables deep-sea deployment.
- Modular design: Sensors can be calibrated/replaced independently.

KEY CHALLENGES

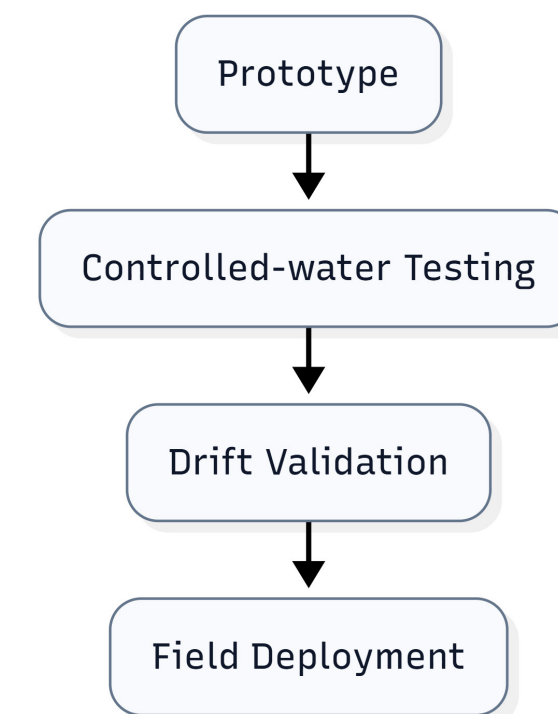
- Underwater drift: Deployment GPS $\neq$ actual seabed sensor position.
- Magnetic interference: PI coil, battery and magnetometer can interfere with each other.
- Accelerometer drift: IMU cannot reliably estimate the full 400 m displacement by double integration.
- Deep-sea conditions: Pressure, waterproofing and seawater exposure.
- PI coil optimization: Final coil design and detection performance require prototype testing.

MITIGATION STRATEGY

- Drift $\rightarrow$ Estimate probable drift direction using RM3100 + MPU6050 + mechanical fins.
- Magnetic interference $\rightarrow$ Physically separate magnetometer and PI coil + time-multiplex operation.
- IMU error $\rightarrow$ Use IMU for short-term motion, not long-term dead reckoning.
- Water/pressure $\rightarrow$ Sealed pressure-resistant housing + protected seawater interface for PT100.
- Uncertainty $\rightarrow$ Represent location as a directional sector, validated through prototype drift tests.

WHY THIS IS VIALE

VALIDATION ROADMAP



LOW-COST HARDWARE
ESP32 + COTS Sensors

SENSOR FUSION
Multi-sensor Evidence +
Drift Estimation

ACTIONABLE OUTPUT
Narrower Search Region $\rightarrow$
Targeted ROV Investigation

1. FASTER EXPLORATION

- Rapid screening of seabed anomalies
- Helps prioritize promising regions before detailed investigation
- Reduces time spent investigating low-value anomalies

2. COST SAVINGS

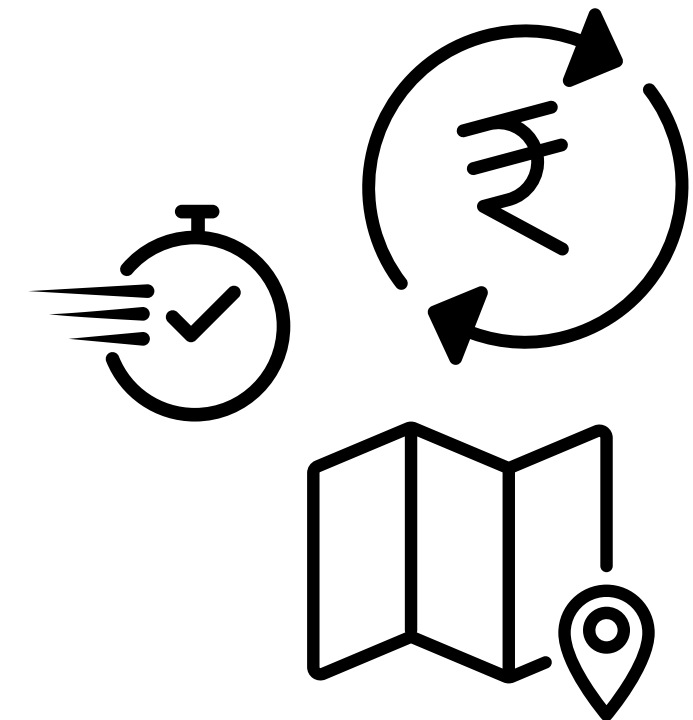
- Low-cost deployable sensor platform
- Reduces unnecessary ROV investigations
- Enables multiple screening deployments during a survey

3. DEEP-SEA RESOURCE MAPPING

- Identifies magnetic, metallic and thermal anomalies
- Combines sensor readings with existing seafloor topography
- Produces spatially constrained candidate regions

4. STRATEGIC BENEFIT

- Supports Indian deep-ocean resource exploration
- Provides an indigenous, modular sensing approach
- Can complement existing MoES/NCPOR survey infrastructure



DETECT

PRIORITIZE

INVESTIGATE

Research:

- [Drone Flow User Guide](#)
- [A Low-Cost and High-Precision Underwater Integrated Navigation System](#)
- [A Multi-Sensor Fusion Underwater Localization Method Based on Unscented Kalman Filter on Manifolds](#)

Hardware References:

- [ESP32-WROOM-32E Datasheet](#)
- [RM3100 Datasheet](#)
- [MPU6050 Datasheet](#)
- [MAX31865 Datasheet](#)
- [DS3231 Datasheet](#)
- [PT100 Datasheet](#)
- [PI Detection Reference](#)

Implementation & Documentation:

- [GitHub Repository](#)